EXAMPLE 4.19

The equation for an alternating current is given by $i = 77\ sin\ 314t$. Find the peak current, frequency, time period and instantaneous value of current at $t = 2$ ms.

Solution

$i = 77 \sin 314\ t$; $t = 2$ m s $= 2 \times 10^{-3}$ s

The general equation of an alternating current is $i = I_m \sin \omega t$. On comparison,

(i) Peak current, $I_m = 77$ A

(ii) Frequency, $f = \dfrac{\omega}{2\pi} = \dfrac{314}{2 \times 3.14} = 50$ Hz

(iii) Time period, $T = \dfrac{1}{f} = \dfrac{1}{50} = 0.02$ s

(iv) At $t = 2$ m s, Instantaneous current,

$$i = 77 \sin(314 \times 2 \times 10^{-3})$$

$$= 77 \sin\left(314 \times 2 \times 10^{-3} \times \frac{180^\circ}{3.14}\right)$$

$$= 77 \sin 36^\circ = 77 \times 0.5878$$

$$= 45.26 \text{A}$$

Phasor and phasor diagram

Phasor

A sinusoidal alternating voltage (or current) can be represented by a vector which rotates about the origin in anti-clockwise direction at a constant angular velocity ω. Such a rotating vector is called a phasor. A phasor is drawn in such a way that

- the length of the line segment equals the peak value V_m (or I_m) of the alternating voltage (or current)
- its angular velocity ω is equal to the angular frequency of the alternating voltage (or current)
- the projection of phasor on any vertical axis gives the instantaneous value of the alternating voltage (or current)
- the angle between the phasor and the axis of reference (positive x-axis) indicates the phase of the alternating voltage (or current).

The notion of phasors is introduced to analyse phase relationship between voltage and current in different AC circuits.

Phasor diagram

The diagram which shows various phasors and their phase relations is called phasor diagram. Consider a sinusoidal alternating voltage $v = V_m \sin \omega t$ applied to a circuit. This voltage can be represented by a phasor, namely $\overrightarrow{OA}$ as shown in Figure 4.38.

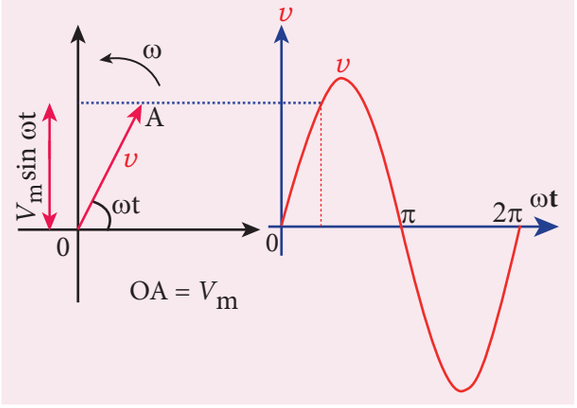


Figure 4.38 Phasor diagram for an alternating voltage $v = V_m \sin \omega t$

Here the length of $\overrightarrow{OA}$ equals the peak value (V_m), the angle it makes with x-axis gives the phase (ωt) of the applied voltage. Its projection on y-axis provides the instantaneous value ($V_m \sin \omega t$) at that instant.

When $\overrightarrow{OA}$ rotates about O with angular velocity ω in anti-clockwise direction, the waveform of the voltage is generated. For one full rotation of $\overrightarrow{OA}$, one cycle of voltage is produced.

The alternating current in the same circuit may be given by the relation $i = I_m \sin\ (\omega t + \phi)$ which is represented by

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